1.1 Machine Learning Basics

Supervised, unsupervised, batch versus online, and the usual failure modes.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

The figures follow Géron, *Hands-On Machine Learning*, Ch. 1. *Mathematics for Machine Learning* is the math source for the rest of Module 1.

1. Types of machine learning systems

The algorithms in this course change. The **setup** usually does not.

Supervised. The training rows come with answers (labels).

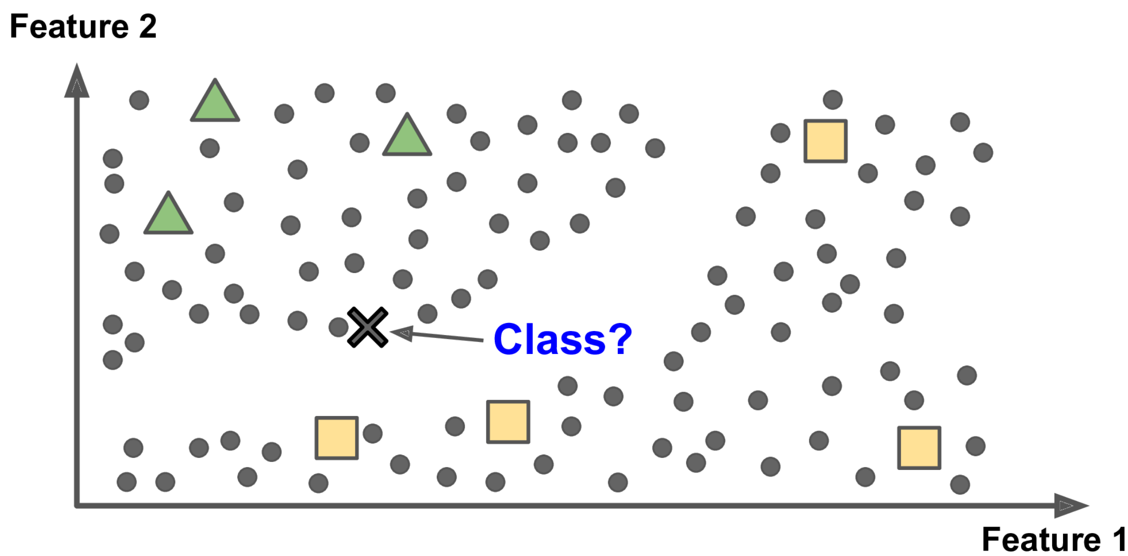
- **Classification** puts a row in a bucket (spam / ham).
- **Regression** predicts a number (price of a car).

Methods you already know still apply: k-NN, linear and logistic regression, SVMs, trees, forests, neural nets. Weeks 5 and 12–15 pick these up again with more math.

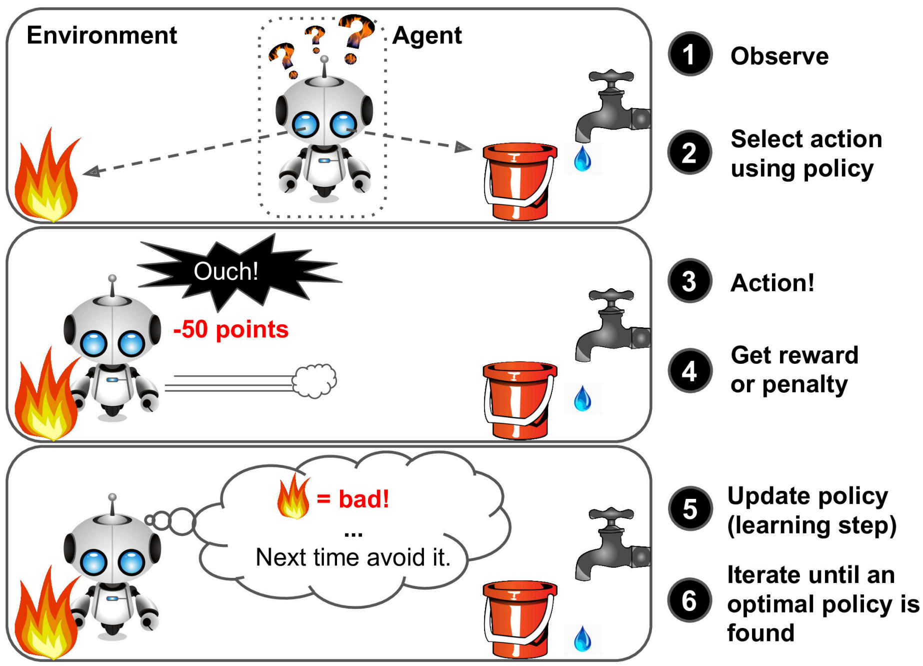
Unsupervised. No labels.

Family	Examples in this course
Clustering	k-means, hierarchical
Anomaly / novelty	one-class SVM; later GMMs
Latent variables	PCA, kernel PCA, ICA, tensors
Association rules	Apriori, Eclat (know the names)

Semi-supervised. A little labeled data, a lot of unlabeled. A photo app clusters faces, then asks you to name each cluster. One label per person names everyone in the album.



Reinforcement. An **agent** takes actions, sees a reward (or a penalty), and updates a **policy**. We will not live here. Know the words.



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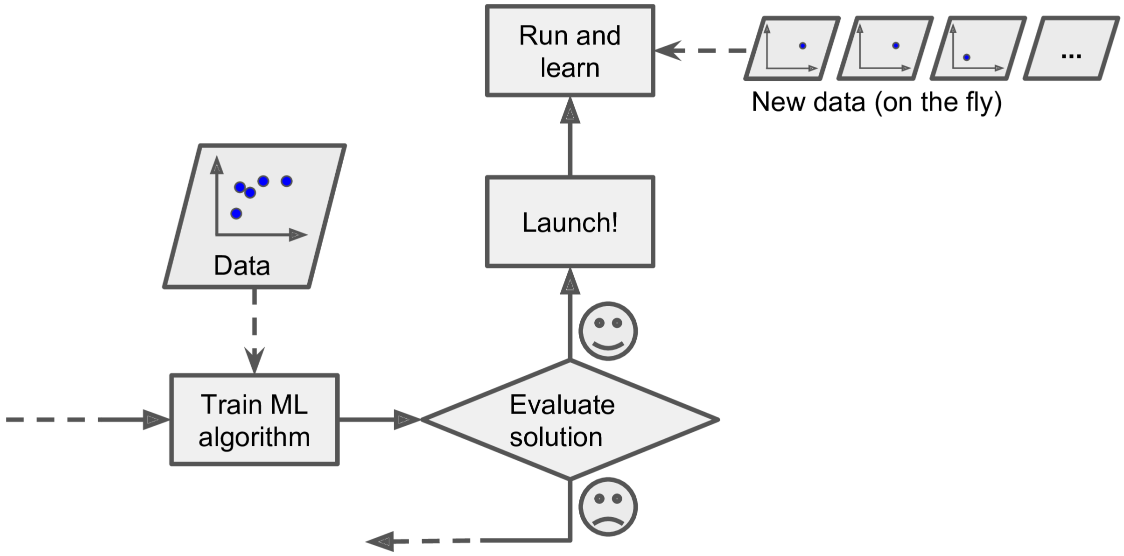
2. Batch versus online

Another cut: can the system learn from a **stream**, or must it see the whole matrix at once?

Batch (offline). Train on all available data, then freeze the model. You can still adapt: collect new data, retrain from scratch, ship a new version. Simple. Slow if the matrix is huge. Fine when the world changes on a daily or weekly clock. Bad for a phone, a rover, or a stock ticker.

Online. Update as points arrive.



The **learning rate** is how fast you forget the past.

- High: you track a changing stream, and you also chase noise.
- Low: you are stable, and you are late.

Online learning does not need the whole dataset in RAM. It can also be order-sensitive and poisoned by outliers. A live system that trains online can be wrecked by one bad sensor or by someone stuffing a search ranking. Monitor the inputs. Be ready to switch learning off and roll back.

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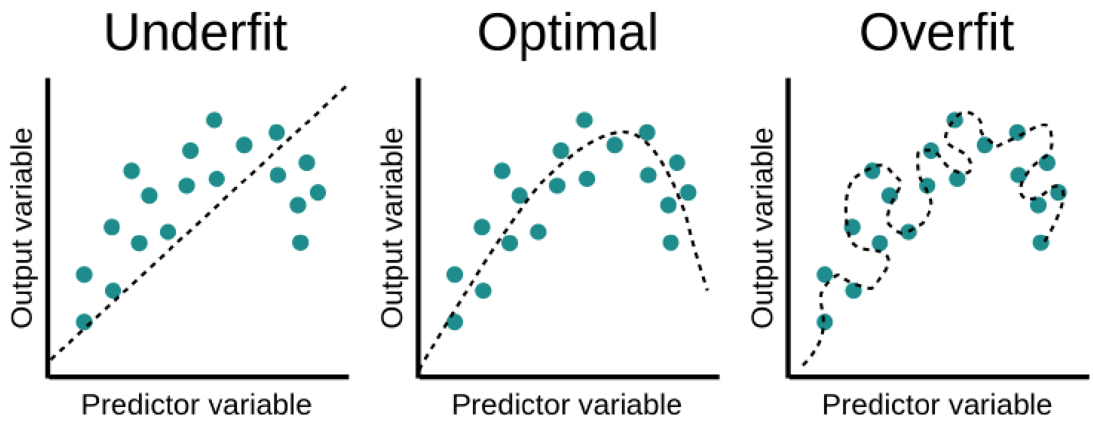
3. Why models fail

Problem	What it looks like	What you do
Too little data	High variance, lucky splits	More data, or a simpler model. Medium data is still the usual case here — do not abandon the math.
Non-representative data	Sampling noise or sampling **bias**; the test world is not the train world	Fix how you sample
Poor quality	Errors, outliers, sensor junk	Clean before you model
Irrelevant features	The signal is not in the columns	Select, extract, or collect better ones
Underfitting	Train and test both bad	Richer model, better features, less regularization
Overfitting	Train great, test poor	Simpler model, more data, less noise



4. Practice

1. Name one supervised and one unsupervised method this course will treat later, and what label each one does or does not see.
2. A spam filter is retrained every night on the last 30 days of mail. Is that batch or online? What breaks if the stream is attacked for one afternoon?
3. Why can a very large training set still fail to represent the world you care about?